

The Sequencing Experiment Design Advisor

Evidence from public archives, measured coverage and transparent budget decisions

Zhaulybaeva Gaukhar
Kurmangazykyzy Dilyara

Astana IT University
Introduction to Bioinformatics
6 October 2026

Measured headline

The project analysed 800,000 Illumina reads and 60,000 Nanopore reads against the 4,641,652-bp MG1655 reference. The completed metadata cohort contains 999,988 archive records. Fixed-reference alignment produced depth-dependent coverage curves rather than an assumed universal depth rule.

Measured endpoint	Illumina	Nanopore
Primary mapped reads	99.60%	99.98%
Accepted-molecule raw nominal depth	40.35x	45.05x
Reference breadth at >=1x	99.14%	99.94%
Reference-breadth plateau	14.12x	15.81x

The runs come from different studies and BioSamples. These measurements demonstrate reproducible reference coverage, with no claim of isolated platform superiority or validated variant-caller sensitivity. The advisor combines this limited evidence with declared planning priors, sampling equations and editable cost assumptions.

Biological question and design limits

A research group must choose a sequencing design that can answer its biological question within a finite budget. We ask: how much additional reference support does extra sequencing deliver in real bacterial short-read and long-read datasets, and how can that evidence constrain an explainable design recommendation?

Application	What adequate data must support	Why a depth rule differs
Variant detection	Several credible alternate observations at an accessible locus	Allele fraction, mapping ambiguity, base errors and caller specificity determine whether a variant is recoverable.
Expression quantification	Counts across biological replicates and transcript abundance	Read-pair count, library composition and replication drive uncertainty; genome coverage is the wrong primary denominator.
Assembly	Overlapping reads that distinguish genome paths	Repeat length and molecule span matter. Increasing short-read depth cannot identify a repeat path when all molecules are too short.

Platform implications

Illumina provides short paired reads and economical SNV evidence in accessible regions, but substitutions, adapters, declining tail quality and amplification bias require QC. PacBio HiFi uses circular consensus to improve base accuracy with long molecules. Nanopore measures single molecules and offers variable read lengths; extraction, chemistry and basecalling strongly affect the distribution and residual error profile. These characteristics change the reachable biological endpoint, not only the file size.

Long reads can be necessary when the question demands a continuous molecule spanning a repeat or an entire transcript isoform and the available short-read fragments provide no distinguishing linkage. That is a conditional identifiability requirement. It does not establish that every assembly or expression experiment needs long reads.

Success definition

The executed benchmark measures reference breadth at $\geq 1x$, $\geq 5x$ and $\geq 10x$, primary mapping rate and alignment-quality proxies as retained depth changes. A coverage plateau is an endpoint-specific finding. It supplies a lower-level coverage diagnostic, while downstream SNV, expression and assembly accuracy remain separate validation questions.

Data sources, identifiers and provenance

Source	Archive identifiers	Acquisition
Illumina MiSeq	SRR1030394; SAMN02401358; PRJNA227741	400,000 paired molecules; archived WGS
Nanopore GridION	SRR33602302; SRX28830916; SAMN48541625; PRJNA1222438 / SRP563276	60,000 single reads; archived WGA label
Reference genome	GenBank U00096.3; RefSeq counterpart NC_000913.3	Complete MG1655 reference, 4,641,652 bp
Metadata cohort	ENA Portal read_run records; release-year query 2015-2025	999,988 processed records

Retrieval scripts resolve run accessions to BioSample and BioProject records and obtain FASTQ locations and archive-reported sizes. Source URLs, retrieval time and SHA-256 of acquired files are saved in `acquisition_manifest.json`. ENA and NCBI mirror INSDC records; they are not counted as two independent datasets. The reference is a sequence accession, rather than a project master record without sequence.

Detected inconsistencies

The Nanopore record says Design: WGS in the NCBI overview but Library strategy: WGA in the structured NCBI and ENA fields. Amplification status therefore remains uncertain. ENA `first_public` is 2025-05-24, one day later than the NCBI publication date 2025-05-23. We preserve both meanings rather than silently forcing them to agree.

For the selected paired Illumina run, `base_count/read_count` is approximately 486 bp, which behaves as a pair/spot span, while the individual FASTQ reads average about 243 bp. Metadata aggregates use the conservative label bases per archive count unit. They do not silently substitute spots for reads or infer a length distribution from a run average.

In the completed cohort, 1,482 records lacked organism names and 31 lacked BioSample identifiers. All records had a run accession and platform. The query did not request a spot-count column, so its absence is a schema limitation rather than a missing-value rate for archived experiments.

Acquisition scope

To keep the experiment tractable, acquisition stops after deterministic archive prefixes. Nested random hash subsamples are then drawn within those prefixes. The initial acquisition is ordered and may contain lane/time/quality bias. Original source-file checksums cannot validate an incomplete prefix; local SHA-256 verifies the acquired artifact only. The two runs share the named strain but not the BioSample, study, instrument generation or library annotation.

Methods: QC, alignment and sampling

The pipeline parses FASTQ as a stream, requires complete four-line records, checks sequence/quality length and DNA symbols, and preserves paired molecule IDs. Short and long reads use different filtering rules. Raw and post-filter QC expose read counts, length quantiles, GC, ambiguous bases, Q20/Q30 fractions and Phred-based expected error. The latter is a calibration-dependent expectation, not truth against an independent genome.

Rule	Reason and practical limit
Illumina: Q20 tail trimming and adapter-prefix audit	Phred Q20 corresponds to 1% nominal base-error probability. A fixed 12-base adapter prefix is a screening signal, not a complete adapter classifier.
Illumina: ≥ 50 bp, $\leq 2\%$ N, mean Phred ≥ 20	Keeps enough seedable sequence while removing highly ambiguous molecules. These are declared conservative educational thresholds, not optimized universal cutoffs.
Nanopore: ≥ 500 bp, $\leq 2\%$ N, mean Phred ≥ 7	Retains noisy long molecules rather than applying Illumina assumptions. Q7 alone is permissive; mapping and alignment metrics provide additional checks.
Coverage: primary M/=X bases, MAPQ ≥ 20	Nominal mapping error at Q20 is 1% only if the aligner score is calibrated. Score255 is unavailable and excluded. D/N gaps and secondary/supplementary hits do not add aligned-base coverage.

Executed alignment uses minimap2 2.22-r1101 via the official Biowasm SIMD build, with sr for paired Illumina and map-ont for Nanopore. Minimapp2 finds minimizers, chains collinear seed hits and performs base-level alignment extension. Distinct seed/scoring presets match the read type. Standard SAM CIGAR operations establish measured aligned intervals; the educational anchor mapper is not the source of delivered headline curves.

A stable seeded hash of normalized molecule IDs creates nested fractions across three seeds. Mates share one hash. Reusing a fixed SAM alignment is valid for the measured fixed-reference per-read/pair endpoint: removing other molecules does not alter those alignments. A supplied native Snakemake workflow realigns each subset. In the executed route, QC selects molecules but their raw-read SAM intervals retain any trimmed tails; this scope is explicit.

Results: quality and data retention

Metric	Illumina	Nanopore
Raw reads	800,000	60,000
Raw bases	194,530,654	212,067,909
Mean read length, bp	243.2	3534.5
Median read length, bp	251	1,848
90th-percentile length, bp	251	8,660
Mean Phred	32.76	39.98
Bases with Q >=20	84.74%	94.58%
Bases with Q >=30	78.50%	89.39%
GC among A/C/G/T	51.06%	51.59%
Ambiguous-base fraction	0.00%	0.00%
Adapter-prefix read fraction	0.00%	0.00%
Retained molecules	384,691	50,175
Excluded molecules	15,309	9,825

Table 1. Measurements from the acquired prefixes. Illumina read count includes both mates. Retained nominal depth uses original bases of accepted molecules in the raw-SAM route. QC thresholds are declared in the methods and every rejection reason is available in summary.json. Identical-sequence duplication estimates use a bounded first-100,000-sequence sample and cannot distinguish PCR duplication from biological repetition.

Observed mean alignment edit distance per aligned query base is 0.0095 for Illumina and 0.0061 for Nanopore. This NM-derived proxy combines sequencing mistakes, true biological differences and alignment behavior. It should not be reported as a calibrated platform base-error rate.

Short-read per-cycle quality and 10-kb reference-window GC/dropout records provide inspectable evidence of quality decay and uneven coverage. The pilot does not mark duplicates or correct overlapping mates, so its read depth differs from independent-molecule depth. It also does not independently prove that all acquired DNA matches the reference strain.

Results: empirical saturation curves

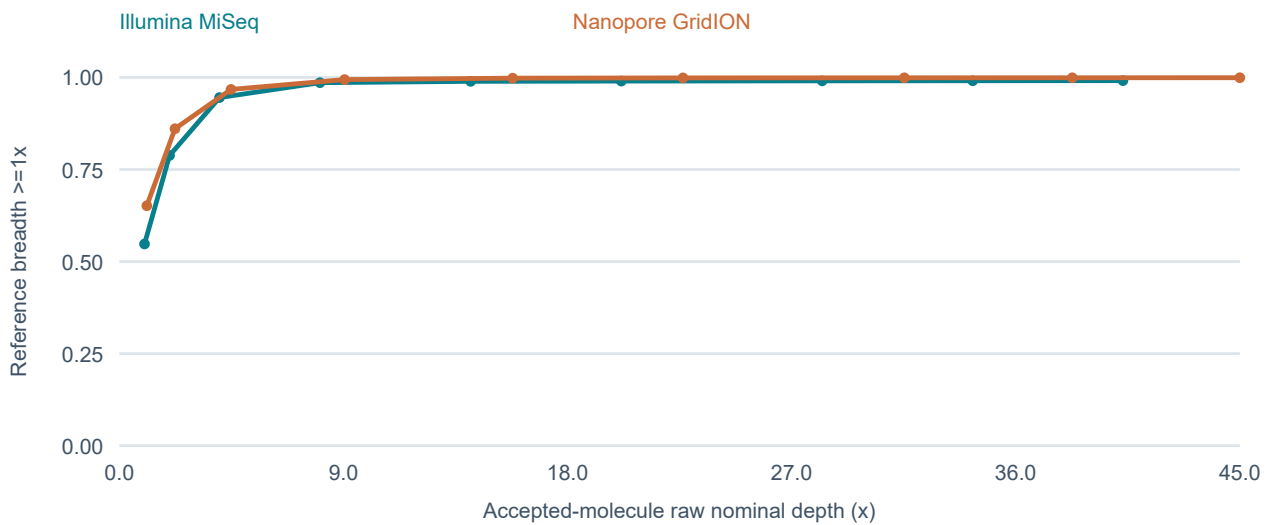


Figure 1. Means across three nested seeded subsamples. Values come from primary minimap2 SAM alignment bases passing the coverage filters. Replicate minima/maxima are retained in saturation.json; the seeds reuse the same acquired run prefix and are not independent biological replicates.

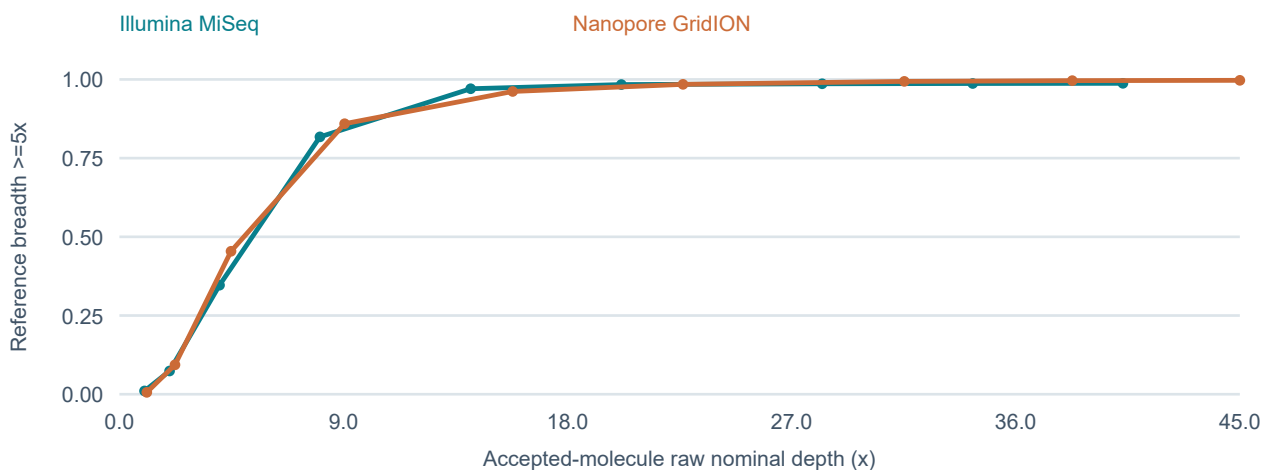


Figure 2. The stricter $\geq 5x$ endpoint saturates later than merely touching a reference position. $\geq 10x$ curves are also supplied. A single depth cannot represent all three requirements.

The predeclared $\geq 1x$ plateau criterion is at least 95% breadth and two consecutive adjacent absolute gains below one percentage point. It detected a plateau at 14.12x for Illumina and 15.81x for Nanopore. The criterion depends on the chosen fraction grid and observable endpoint. Coverage beyond a plateau can still improve alternate-read support or harder-region depth.

Results: mapping, quality and archive designs

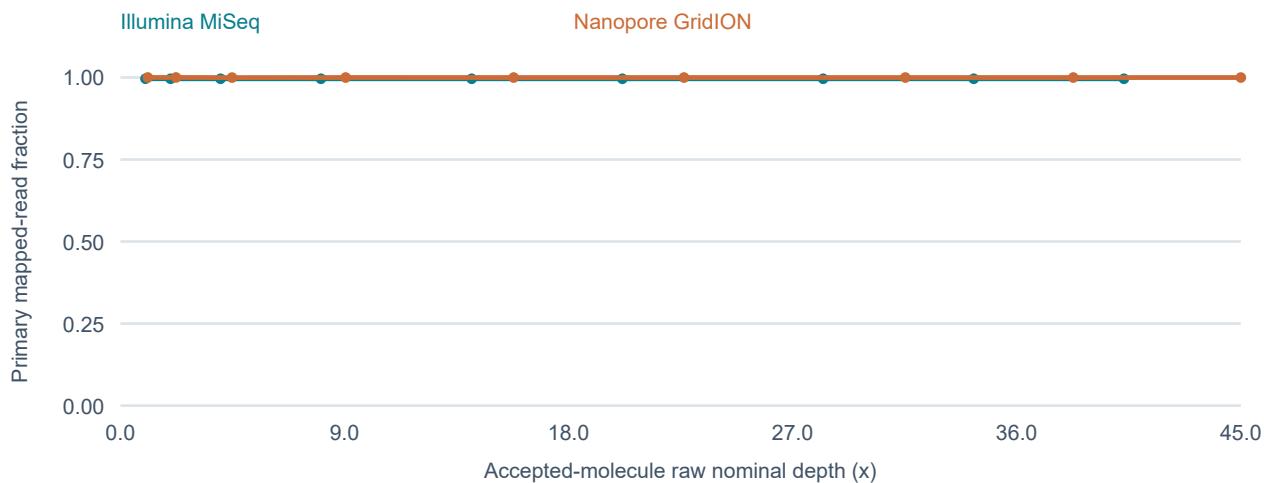


Figure 3. Mapping is largely insensitive to the number of other reads when reference and aligner settings are fixed. Sampling composition causes small fluctuations. This is why coverage, rather than an assumed declining mapping fraction, constrains the depth experiment.

Quality across all measured fractions	Illumina	Nanopore
Mean MAPQ	58.2581 to 58.3329	59.7702 to 59.8141
Mean NM / aligned-query bases	0.0095 to 0.0095	0.0059 to 0.0062

The metadata analysis streamed 999,988 records and grouped release year, platform, library strategy and layout. Platform counts in this query cohort are: ILLUMINA: 910,759; PACBIO_SMRT: 16,233; OXFORD_NANOPORE: 43,686; ION_TORRENT: 11,511; DNBSEQ: 13,467; LS454: 217; BGISEQ: 2,991; CAPILLARY: 694; ABI_SOLID: 205; TAPESTRI: 10; ELEMENT: 214; COMPLETE_GENOMICS: 1.

Year	Platform	Strategy	Runs
2023	ILLUMINA	WGS	157,833
2024	ILLUMINA	WGS	113,503
2023	ILLUMINA	AMPLICON	79,477
2025	ILLUMINA	AMPLICON	66,148

Year means public archive release, not sequencing date. The bounded, unsorted query over-represents whichever records the API returns first. Run counts are not study counts, biological sample counts, technology market share or evidence that a design succeeded. Strategy labels are coarse application proxies. The cross-archive sample/project joins for the selected runs supply curated context.

Cost model and recommendation cases

The estimate sums library preparation, sample QC, sequencing consumables, instrument allocation, analysis compute, retained storage and 15% contingency. Raw yield divides the effective requirement by declared usable and mapped fractions. Shared procurement allocates capacity but includes minimum sequencing charges; dedicated procurement rounds up to whole flow cells. Extraction, shipping, taxes, some labor and institutional overhead remain outside the model.

Scenario	Returned design	Estimated USD	Budget outcome
6 bacterial isolates, clonal SNVs	Illumina short reads; 20x	748.17	within_estimate
1 human, 5% allele	Illumina short reads; 126x	3086.18	over_budget
Bacterial repeat resolution	Oxford Nanopore; 40x	220.89	within_estimate
6 RNA samples, 2 conditions	Illumina short reads; fragment target	1584.11	within_estimate

Predicted detection support

For ideal independent accessible-locus sampling, K follows Poisson($D \times f$), where D is effective depth and f is allele fraction. The advisor solves $P(K \geq k)$ for the requested confidence. For $k=3$ and 95%, the required mean alternate count is 6.2958. A 5% allele therefore needs approximately 126x effective mean depth for sampling support alone. Sequencing error, strand balance, locus accessibility and caller specificity are not validated by that calculation.

Published cost checks

Vendor reagent prices are kept separate from external service prices. An independent Northwestern NUSeq FY26 25B PE150 lane costs \$3,750 for 750-900 Gb, or \$4.167-\$5/Gb. The model sequencing-plus-instrument allocation is \$5.15/Gb: 3%-23.6% above that range. A 10B lane has a different \$5.205-\$6.246/Gb envelope. The model DNA preparation assumption of \$80/sample is 66.7% below the listed \$240 service charge, so project totals can substantially understate a facility order.

The verified \$840 single Nanopore DNA flow-cell price excludes preparation and instrument cost. Compute \$0.35/hour and storage \$0.023/GB-month are exposed planning assumptions, not current quotes for a chosen region. The displayed +/-30% sensitivity band is not a statistical confidence interval. A facility-specific quote should replace assumptions before an order.

Engineering, reproducibility and limitations

The executed Python analysis took 244.7 seconds and recorded peak process memory 748.3 MiB. The coverage engine was `numpy_difference_arrays`. Minimap2 SIMD short-read alignment took 181.35 seconds and long-read alignment 221.93 seconds on this host. These timings exclude failed scalar experiments, acquisition, and the Python analysis stage.

Streaming metadata aggregation took 50.40 seconds at 19,841 records/second and peak memory 36.82 MiB. Fresh-process 1,000/10,000/100,000-record prefixes and the full cohort appear in `metadata_scale.json`. Prefixes test computational scale and are not probability samples of archive designs.

FASTQ QC and metadata ingestion stream records rather than loading raw files. Compact JSON/CSV aggregates belong in Git, while raw FASTQ and SAM stay excluded. The coverage pilot retains per-molecule alignments in memory and is a bounded bacterial workload. The optional native Snakemake workflow uses coordinate-sorted BAM and samtools depth, with per-rule runtime/memory benchmarks. That native route and the Docker image were supplied but not executed on the Windows host.

One-command verification

`python run_experiment.py` runs the included small real fixture through checksum-pinned minimap2 and the analysis. `python run_experiment.py --pilot` retrieves and reproduces the headline bounded raw-read experiment. The source README specifies Python, Node.js and optional NumPy versions. GitHub Actions runs unit tests and the real-read quick workflow after upload. Local tests cover malformed FASTQ, paired identity, nested sampling, CIGAR boundaries, reverse/repetitive alignments, coverage metrics, input validation, Poisson support and procurement arithmetic.

Limits on the conclusions

The prefixes may be biased, the platform comparison is confounded by sample/study/year, and the ONT library annotation is inconsistent. A single reference cannot reveal all strain variation or plasmid content. MAPQ lacks independent calibration, duplicate handling is limited, and QC-selected raw-SAM intervals differ from realigned trimmed reads. Three seeds measure subset variability only. Metadata cannot link success to a design without curated outcome labels. Saturation of reference breadth does not establish downstream accuracy. Cost totals are transparent scenarios rather than purchase quotations.

Conclusion, team responsibilities and references

The advisor makes a practical design explainable: a biological question sets the capability requirement, measured bacterial coverage shows diminishing gains, an explicit sampling model determines alternate-read support, and a full cost breakdown exposes budget assumptions. A recommendation can be feasible under a model while requiring a local service quote or downstream performance validation. The measured curves and negative findings make those limits visible.

Team contributions and allocation of responsibilities

The project divides technical responsibility between the two team members as follows. These roles specify the allocation of tasks; individual completion is documented through the members' own checks, execution records and repository changes.

Team member	Allocated responsibilities
Zhaulybaeva Gaukhar	Dataset selection and accession review; metadata mining; short- and long-read QC; alignment and nested downsampling; coverage and alignment-quality interpretation.
Kurmangazykyzy Dilyara	Platform and cost-source review; sampling and budget model; advisor interface; test scenarios and reproducibility bundle; report and presentation organization.

Both members share responsibility for biological framing, cross-review of results, final scientific verification and presentation of the project. Each member should be able to explain the complete workflow.

AI disclosure

OpenAI Codex substantially assisted requirement extraction, source research, implementation, debugging, tests, real-data processing, interpretation, report and slides. All measured numeric results came from executed archive reads/metadata; model predictions remain labeled. The students are responsible for reviewing the work and explaining submitted components. The repository documents the task allocation and AI assistance.

Selected primary references

[Li \(2018\), Minimap2: pairwise alignment for nucleotide sequences](#)

[ENA Portal: programmatic advanced search](#)

[NCBI, selected ONT experiment SRX28830916](#)

[GenBank reference U00096.3](#)

[Biowasm official project and compiled tools](#)

[Northwestern NUSeq, FY26 service pricing](#)

[Oxford Nanopore US flow-cell store](#)

Additional vendor capability sources, study context and source-check dates are listed in docs/SCIENTIFIC_SOURCES.md and config/costs.json. Source acquisition occurred on 6 October 2026. Course requirements follow AITU_Bioinformatics_Projects-3.pdf, Project 15 and the 60-point rubric.